Assignment 3 · Ground Truth

Alewife Brook Reservation, Cambridge MA · the bank of the Little River beside the constructed stormwater wetland (2013)

Group: Fernanda Castillo, Arturo Gomez Escobedo, Marko Kihara Manriquez

Plant catalog: Fernanda Castillo. Scan, splat, drawings, layout: Marko Kihara Manriquez.

Field visits Oct 2026 · pin-up Fri Oct 30 2026, 12 pm

SHEETS

PIN-01 Part A · Patch Delineation (group delineation)
PIN-02 Part B · Ground-Truthed Patch Map
PIN-03 Part C · Plot + Transect
PIN-04 Part C · Plants
PIN-05 Part C · Field Notes + Observations
PIN-06 Site Photos
PIN-07 Nest · Habitat Feature
PIN-08 Scan + Splat Renders

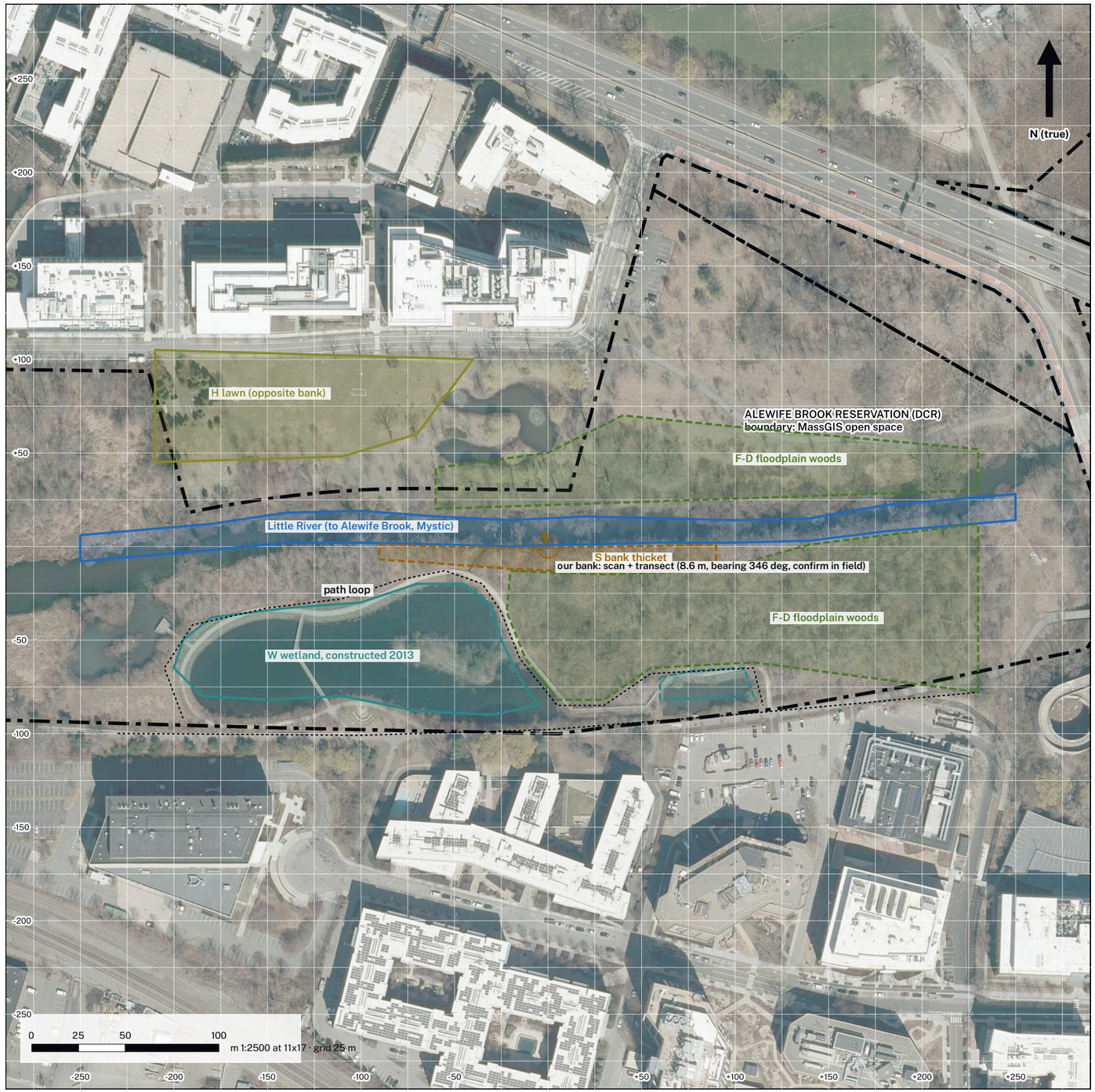
P = estimated, not measured. Grey italic notes = confirm in field. Plant species: Plant catalog: Fernanda Castillo; P1 oak: Marko Kihara Manriquez.

Abandoned bird's nest with woven plastic wrappers, found on the bank (PIN-07).

Part A · Patch Delineation (group delineation)

Group delineation on the MassGIS 2023 leaf-off aerial at 1:2500, from Fernanda Castillo's plant catalog and our field observations (Oct 2026). Solid = edge seen on the aerial and on site; dashed = uncertain.

PIN-01



KEY group delineation (spec A1 classes)

O	open water: Little River
W	wetland: constructed stormwater wetland (2013)
S	shrub: bank thicket (brambles, buckthorn, knotweed)
F-D	woody, deciduous: floodplain woods
H	upland herbaceous: manicured lawn (opposite bank)
path	path: Alewife Brook Pathway loop

solid edge = seen on the aerial and on site · dashed edge = uncertain

Not found in our area: F-E evergreen, B bare ground (except the compacted bank), P paving (outside the reservation).

METHOD

Edges traced by eye on the leaf-off aerial where tone or texture changes over about 10 m (minimum patch about 100 m2), then checked against what we saw on the bank (Oct 2026) and the species in Fernanda Castillo's catalog. Fernanda Castillo's catalog lists species, not positions: knotweed and buckthorn are placed in the bank thicket and disturbed openings, goldenrod at edges, apple and sugar maple in the floodplain woods (approximate). Drawn by Marko Kihara Manriquez from the group's material.

Patches outside our transect are delineated from the aerial only.

FIELD NOTES (observations, Oct 2026)

· Our bank: the south / southwest-facing bank of the Little River, between the river (flowing to Alewife Brook, then the Mystic) and the Alewife Brook Pathway loop.

· Fallen telephone pole across two oaks: it fell onto one mature oak and caught a second at its Y-branch. Moss on pole and oaks, no wires: likely years ago. The gap let saplings in.

· Soil: compacted bank; sticky mud in the wetland bog; dark silty detritus about 6 in over gray clay at about 1 ft; no rotting smell.

· Water: shallow and still, about 0 to 1 ft at the margin, 2 to 3 ft further out; dead trees and branches break the surface.

· Brambles and bush thickets on the shallow bank and in the wetland. Human use: an encampment, waste, plastic.

· Opposite bank (N / NW): manicured lawn, so possible fertiliser and stormwater runoff into the river.

· Two abandoned bird's nests with plastic wrappers woven in (PIN-07).

· Management: the forebay is cleaned out with a front-end loader; habitat monitoring is infrequent; city and state land, overseen by DCR / DEP.

SITE NOTE (City of Cambridge)

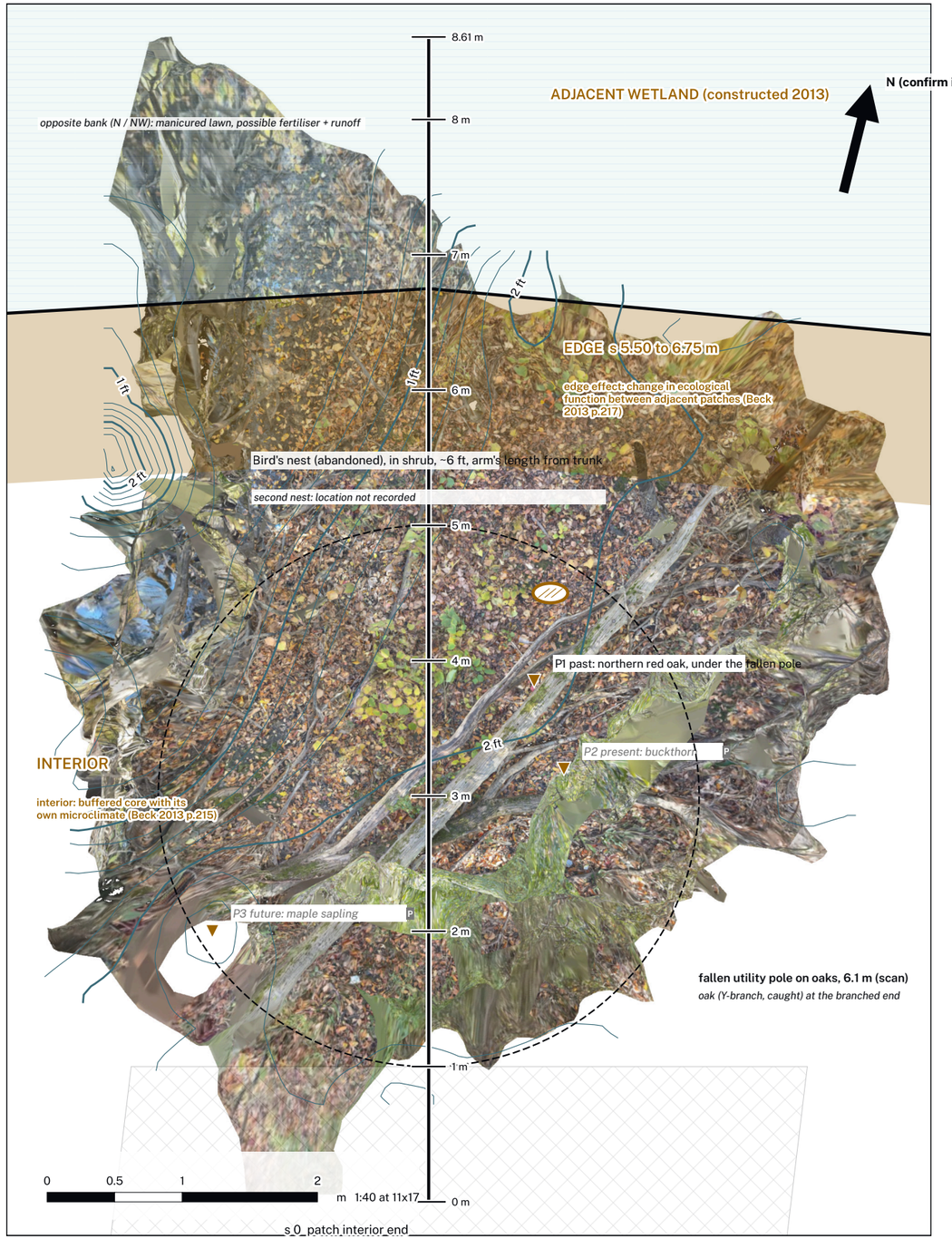
Constructed stormwater wetland, 3.4 ac, opened Oct 15 2013 (City of Cambridge handout p.1 to 2). It attenuates and treats stormwater before it discharges into the Little River (project overview p.1) and enables sewer separation that cuts combined sewer overflows to Alewife Brook by 43.6 million gallons a year (overview p.2). Over 115,000 wetland plants in deep, emergent and high marsh zones (handout p.2). Forebay, basin and oxbow maintained by the City; DCR keeps the recreational parts (overview p.2).

Aerial: MassGIS orthos2023, spring 2023 leaf-off, 15 cm (tiles.arcgis.com/.../orthos2023/MapServer). Boundary: MassGIS Protected and Recreational OpenSpace (gis.eea.mass.gov/server/rest/services/Protected_and_Recreational_OpenSpace_Polygons), Alewife Brook Reservation, DCR. Scan location 42.396871, -71.146994.

Part B · Ground-Truthed Patch Map

Part B. Group patch on the scan footprint: Polycam mesh texture, top view, ortho, plan 1:40 at 11x17. Fallen utility pole on oaks across the plot; species from Fernanda Castillo's plant catalog; marsh zones are the City's planted palette, not observed by us.

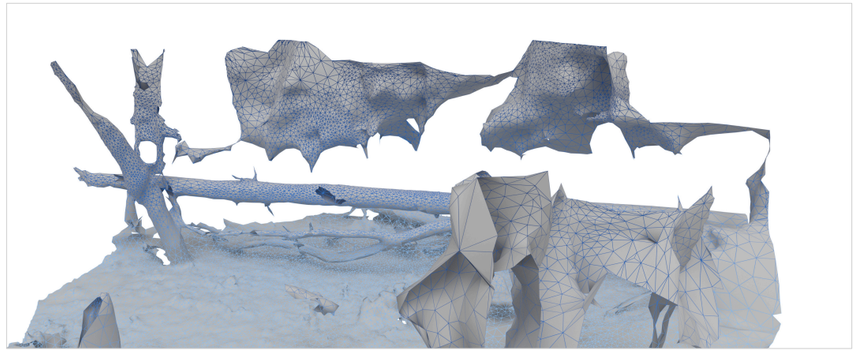
PIN-02



KEY plot layers

- base: Polycam mesh texture, top view, ortho
- ground contours 0.25 ft (scan ground grid)
- index contour 1 ft, labelled (ft, scan z datum; relief 0.1 to 2.8 ft)
- edge: functional band 1.25 m (s 5.50 to 6.75)
- group patch boundary (estimated outline)
- adjacent patches, spec A1 classes (estimated)
- transect, ticks every 1 m, s 0 at the patch interior
- fixed-radius plot r 2.0 m
- P1 past (oak under the fallen pole) · P2 present · P3 future
- habitat feature: bird's nest (abandoned), position: confirm in field

12.3 m measured (scan / field)



Mesh wireframe, pole on oaks (Polycam scan) trimmed mesh, front-angled, 30 mm; wire blue by height

term (reading, page)

estimated P

P = estimated, not measured.

SITE NOTE

Constructed stormwater wetland, 3.4 ac, opened Oct 15 2013 (City of Cambridge handout p.1 to 2). It attenuates and treats stormwater before it discharges into the Little River (project overview p.1) and enables sewer separation that cuts combined sewer overflows to Alewife Brook by 43.6 million gallons a year (overview p.2). Over 115,000 wetland plants in deep, emergent and high marsh zones (handout p.2). Forebay, basin and oxbow maintained by the City; DCR keeps the recreational parts (overview p.2).

OBSERVED (Oct 2026)

Disturbance: fallen utility pole on two oaks (canopy gap, saplings in, legacy wood); brambles and thickets on the bank; soil compaction; human use (an encampment, waste, plastic). Adjacency: manicured lawn on the opposite bank (N / NW), possible fertiliser and stormwater runoff.

TERMS (readings + ett3 lectures)

patch: an area that differs from what surrounds it (Beck 2013 p.211; ett3 lecture, Oct 9 2026)
edge effect: change in ecological function between adjacent patches (Beck 2013 p.217)
disturbance: a process, not just an event: the pole fall opened the canopy (ett3 lecture, Oct 2 2026)
ruderals: fast growers that take disturbed openings (knotweed, buckthorn) (ett3 lecture, Sep 11 2026)

PLANT TABLE primary species: Plant catalog: Fernanda Castillo · companions: oak, pole, nests

common	Latin	form / str.	patch	wetl. ind.	origin / MA	role	credit
Japanese knotweed	*Reynoutria japonica*	herb, tall / U	bank thicket, disturbed edge	FACU	E Asia; invasive, prohibited in MA	edge, disturbance	Plant catalog: Fernanda Castillo
European buckthorn	*Rhamnus cathartica*	shrub / M	canopy gap, bank thicket	FAC	Europe; invasive, prohibited in MA	P2 present	Plant catalog: Fernanda Castillo
Goldenrod	*Solidago spp.*	herb / G	edge, openings	FACU to FACW	native		Plant catalog: Fernanda Castillo
Sugar maple	*Acer saccharum*	tree / O (sapling U)	floodplain woods, gap saplings	FACU	native	P3 future	Plant catalog: Fernanda Castillo
Domestic apple	*Malus domestica*	tree / M	floodplain woods edge	UPL, confirm	Central Asia; non-native, not listed		Plant catalog: Fernanda Castillo
Northern red oak	*Quercus rubra*	tree / O	under the fallen pole	FACU	native	P1 past	Marko Kihara Manriquez
Oak, caught at Y-branch	*Quercus sp.*	tree / O	under the pole		native	companion	observed
Fallen utility pole	*not a plant*	legacy wood / G	across the plot			companion	observed
Bird's nests (2, abandoned)	*habitat feature*	shrub, ~6 ft	s 4 to 5 m; 2nd not recorded			habitat	observed

Indicator: USACE National Wetland Plant List, Northcentral and Northeast. MA invasive: MDAR Prohibited Plant List. Silver maple (A. saccharinum, FACW) is the floodplain maple expected here; confirm against the sugar maple ID. Earlier tentative photo IDs are on REF-03 only.

MARSH ZONATION planted palette per City of Cambridge (designed planting; not observed by us)

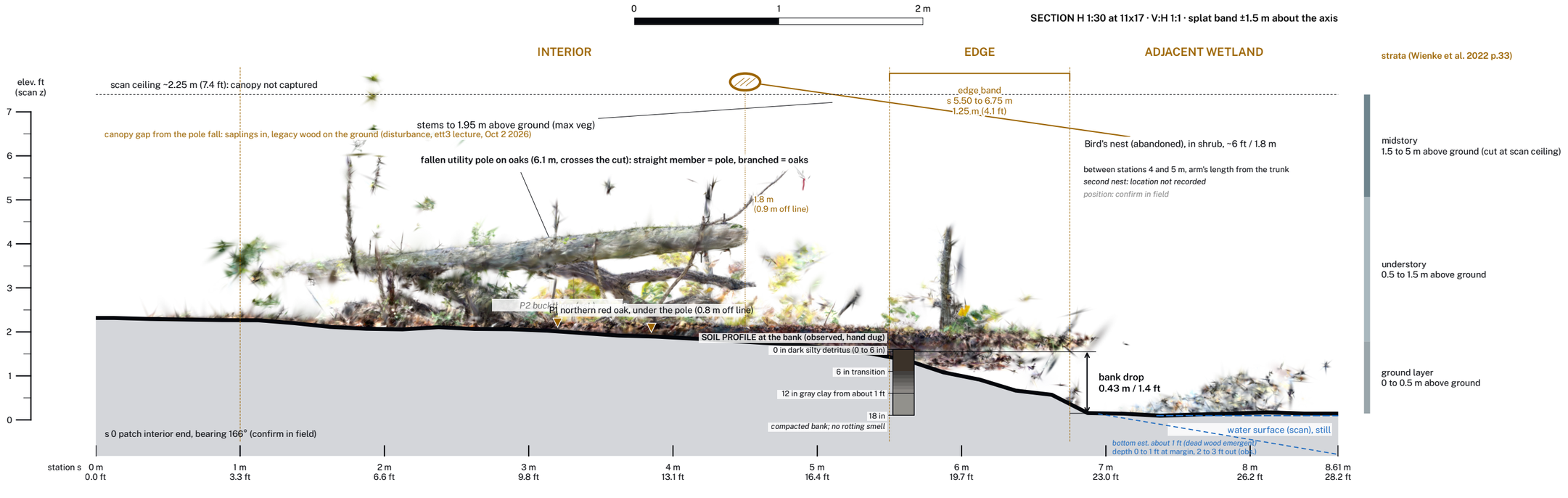
deep marsh	1 to 1.5 ft	hardstem and softstem bulrush, lesser bur-reed
open water	1.5 to 3 ft	white water lily, pickerelweed, northern arrowhead
emergent marsh	6 in to 1 ft	arrow arum, spike rush, green bulrush, blue flag iris
high marsh	up to 6 in	tussock sedge, marsh marigold, riverbank wild rye, Canada rush, wool grass, marsh hibiscus
scrub-shrub	bank	silky dogwood, pussy willow, black chokeberry, sweet pepperbush, meadowsweet, highbush blueberry

Reference gradient for the wetland patch: deep to high marsh, then scrub-shrub at the bank.

Part C · Plot + Transect

PIN-03

Part C composite. Transect section on the gaussian splat side view cut to the transect band (±1.5 m), H 1:30, V:H 1:1, stations every 1 m, s = 0 at the patch interior. Plant table per zone. Plot in plan. Bird nest (habitat feature) projected onto the section.



zone (scan)	gap			interior			edge		adjacent patch	
veg height max, m (scan)	0.31	1.11	1.61	1.47	1.69	1.95	1.07	0.29	0.18	
strata present (scan)	G	G U	G U	G U	G U	G U	G U	G	G	
canopy openness %, densiometer (estimated)	*30* P	*22* P	*18* P	*18* P	*20* P	*25* P	*40* P	*70* P	*80* P	
% cover (estimated)	*35* P	*70* P	*85* P	*80* P	*80* P	*75* P	*60* P	*85* P	*90* P	
no. of species (estimated)	*3* P	*5* P	*6* P	*6* P	*5* P	*6* P	*7* P	*5* P	*4* P	
dominant species (estimated)	*Poa sp.* P	*R. cathartica* P	*R. cathartica* P	*R. cathartica* P	*R. cathartica* P	*A. saccharum* P	*Salix sp.* P	*Impatiens capensis* P	*Carex sp.* P	

PLANT TABLE transect sampling list (spec C2 to C4), one row per zone

zone	dominant species	% cover	no. of species	veg. height	strata present (O / M / U / G)	canopy (shade / partial / open)	woody presence (vines / seedlings / saplings / trees)
interior (s 1 to 5.5 m)	*R. cathartica* P	*71* P	*6* P	1.95 m (6.4 ft)	U G	*shade (22 % open)* P	*saplings, seedlings, vines* P
edge (s 5.5 to 6.75 m)	*A. saccharum* P	*68* P	*7* P	1.07 m (3.5 ft)	U G	*partial (32 % open)* P	*saplings, vines* P
wetland (s 6.75 to 8.61 m)	*Carex sp.* P	*78* P	*7* P	0.91 m (3.0 ft)	U G	*open (63 % open)* P	*seedlings* P

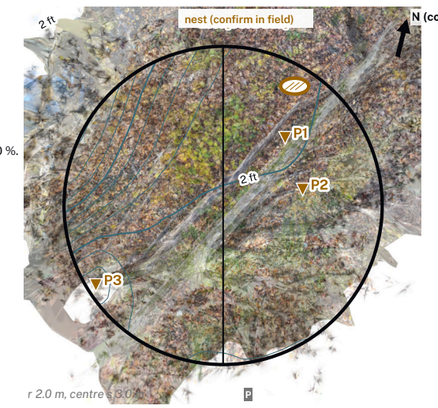
Values in grey italic with P are estimated, not measured. Veg. height and strata from the scan (strata by height above ground; no returns above 2.25 m, so O cannot be read). Canopy from densiometer openness: shade < 25 %, partial 25 to 60 %, open > 60 %.

SPECIES Plant catalog: Fernanda Castillo (primary) · companions: oak (P1, Marko Kihara Manriquez), pole

common	Latin	form / str.	where	ind.	origin / MA	role
Japanese knotweed	*Reynoutria japonica*	herb, tall / U	bank thicket, disturbed edge	FACU	E Asia; invasive, prohibited in MA	edge, disturbance
European buckthorn	*Rhamnus cathartica*	shrub / M	canopy gap, bank thicket	FAC	Europe; invasive, prohibited in MA	P2 present
Goldenrod	*Solidago spp.*	herb / G	edge, openings	FACU to FACW	native	
Sugar maple	*Acer saccharum*	tree / O (sapling U)	floodplain woods, gap saplings	FACU	native	P3 future
Domestic apple	*Malus domestica*	tree / M	floodplain woods edge	UPL, confirm	Central Asia; non-native, not listed	
Northern red oak	*Quercus rubra*	tree / O	under the fallen pole	FACU	native	P1 past
Fallen utility pole	*not a plant*	legacy wood / G	across the plot			companion

Indicator: USACE NWPL, Northcentral and Northeast. MA invasive: MDAR. Silver maple (A. saccharinum) expected on the floodplain; confirm.

PLOT plan 1:75 · splat over mesh, contours 0.25 ft



KEY

- ground section line (scan ground, 5th pct z per 0.25 m)
- below ground (poche)
- water surface (scan), depth observed
- scan ceiling 2.25 m
- edge band, zone breaks

estimated P

P = estimated, not measured.

TERMS (readings)

edge effect: change in ecological function between adjacent patches (Beck 2013 p.217)

interior: buffered core with its own microclimate (Beck 2013 p.215)

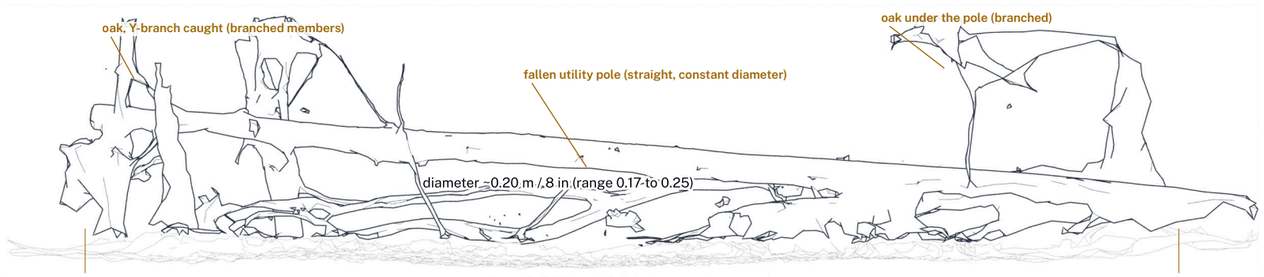
strata: overstory, midstory, understory (Wienke et al. 2022 p.33)

Site: constructed stormwater wetland, 3.4 ac, opened Oct 15 2013; treats stormwater before it reaches the Little River (City of Cambridge handout p.1 to 2, overview p.1).

Part C · Plants

Part C5. Three plants: P1 past, P2 present, P3 future. P1 = northern red oak under the fallen utility pole, line drawing from the scan mesh (Blender Freestyle, ortho side view at 1:30). P2 / P3 per Fernanda Castillo's catalog.

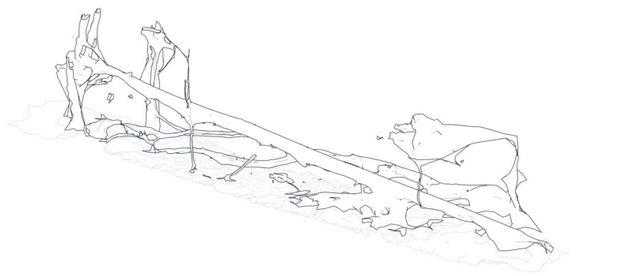
PIN-04



length 5.75 m / 18.9 ft (scan, axis extent)

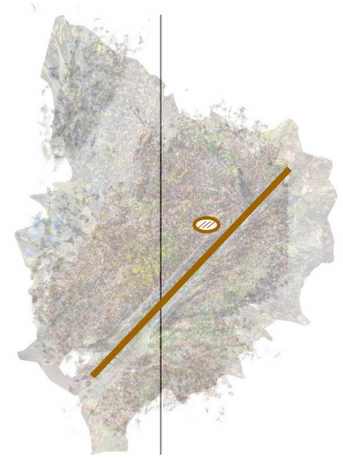
SIDE ELEVATION 1:30 at 11x17 · looking across the log axis · line weights: outline / creases / ground

0 1 2 m



3/4 VIEW not to scale

INSET pole on the site plan (accent), transect, nest



P1 · PAST Northern red oak (Quercus rubra), under the fallen utility pole. ID from form and site, confirm.

Measured (scan): the long straight member of constant diameter is the pole: length 5.75 m (18.9 ft); diameter about 0.20 m (8 in), range 0.17 to 0.25 m; top 0.66 m above ground at the edge end, 0.22 m at the interior end; ground contact about 1.0 m, the rest rests on the caught oak. The curving branched members are the oaks. Not measured in the field: butt diameter, root plate, decay class.

What happened: a telephone pole (no wires, moss on it, so likely years ago) fell onto one mature oak and caught a second oak at its Y-branch, taking it down too.

Now: legacy wood on the ground, moss on the pole and the oaks; a canopy gap overhead with saplings coming in (disturbance-driven succession).

Before: closed oak canopy on the bank. After (expected): the gap fills with saplings; invasive buckthorn and knotweed can take the opening first (hypothesis).

Companion element, the pole: a disturbance agent, not a plant. Wood and preservative treatment speculative; read the brand / tag in the field.

P2 · PRESENT

European buckthorn (Rhamnus cathartica), invasive: colonising the gap

P2 / P3 per Fernanda Castillo's catalog · Plant catalog: Fernanda Castillo

· Introduced as a hedge plant.
· Birds eat and spread the black berries.
· Dense shade suppresses native understory.

Hypothesis: invasives like buckthorn and knotweed thrive in disturbed openings like the pole-fall gap (ruderals, ett3 lecture, Sep 11 2026).

Indicator FAC (NWPL NCNE). Prohibited from sale and distribution in MA (MDAR).

photo: Fernanda Castillo's plant catalog

P3 · FUTURE

Sugar maple sapling (Acer saccharum): canopy return

P2 / P3 per Fernanda Castillo's catalog · Plant catalog: Fernanda Castillo

· Signature New England tree.
· Source of maple syrup.
· Can live 300+ years.
· Food and habitat for birds and squirrels.

Saplings in the gap grow toward the canopy the pole opened (succession, ett3 lecture, Sep 11 2026).

Indicator FACU (NWPL NCNE). Silver maple, A. saccharinum, is the floodplain species expected here; confirm.

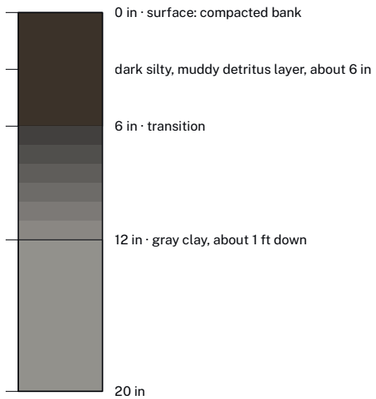
photo: Fernanda Castillo's plant catalog

Part C · Field Notes + Observations

Marko Kihara Manriquez's field observations, Alewife Brook Reservation, Oct 2026, as an annotated sheet. No hand sections were received: one frame is kept for them (a deviation from spec C3, logged).

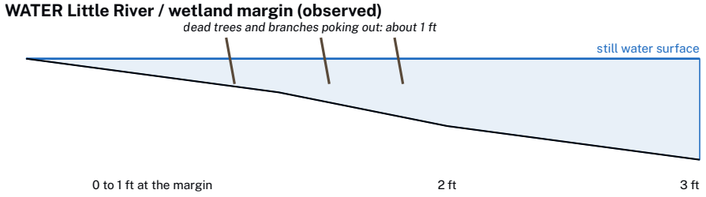
PIN-05

SOIL PROFILE at the bank (observed)



Sticky mud in the wetland bog. No rotting smell detected. Fills the spec soil-horizons item; depths by eye and hand, not measured with a probe.

WATER Little River / wetland margin (observed)



Shallow, still, slow flow from the wetland forebay toward the Little River, then Alewife Brook and the Mystic. Bottom of fine silt, sand and mud (habitat for burrowing worms, snails, aquatic insects, freshwater mussels). Section not to scale.

DISTURBANCES (spec C1)

· Fallen telephone pole on two oaks: onto one mature oak, catching a second at its Y-branch. Moss on both, no wires: likely years ago.
· Canopy gap from that fall: saplings in (disturbance-driven succession).
· Brambles and bush thickets on the shallow bank and in the wetland.
· Soil compaction on the bank.
· Human use: an encampment, human waste, plastic.

NESTS (habitat feature)

· Two abandoned bird's nests with plastic wrappers woven into the structure: deliberate, not blown in. Adaptive reuse of plastic litter, possibly as weatherproofing.
· Nest 1: in a shrub, about arm's length from the trunk, about 6 ft up, between stations 4 and 5 m.
· Nest 2: location not recorded. Both collected (PIN-07).

ADJACENCY

· Our bank faces south / southwest, between the Little River and the Alewife Brook Pathway loop.
· Opposite bank (N / NW): manicured lawn, so possible fertiliser and stormwater runoff.

MANAGEMENT

· The wetland forebay is cleaned out with a front-end loader (the City's term: "objectionable solids").
· The City monitors wetland habitat, but infrequently.
· Ownership: a mix of city and state land, overseen by DCR / DEP.
· Constructed stormwater wetland, 3.4 ac, opened Oct 15 2013 (City of Cambridge handout p.1 to 2).

MARSH ZONATION planted palette per City of Cambridge (not observed by us)

deep marsh (1 to 1.5 ft)	hardstem and softstem bulrush, lesser bur-reed
open water (1.5 to 3 ft)	white water lily, pickerelweed, northern arrowhead
emergent marsh (6 in to 1 ft)	arrow arum, spike rush, green bulrush, blue flag iris
high marsh (up to 6 in)	tussock sedge, marsh marigold, riverbank wild rye, Canada rush, wool grass, marsh hibiscus
scrub-shrub (bank)	silky dogwood, pussy willow, black chokeberry, sweet pepperbush, meadowsweet, highbush blueberry

HAND SECTION (to be added)

hand section of the transect, to be added

s 0 interior — adjacent s 8.61

TERMS ett3 lectures (class notes)

patch: an internally homogeneous area that differs from its surroundings (ett3 lecture, Oct 9 2026)

edge structure: abrupt, feathered, gradual edge; forest interior (ett3 lecture, Oct 9 2026)

disturbance: a force or process causing a perturbation relative to a reference state; a process, not just an event (ett3 lecture, Oct 2 2026)

succession: a continual shifting of conditions within an environment (ett3 lecture, Sep 11 2026)

ruderals (CSR): fast growing, many seeds, germinate in many conditions; non-native is not the same as invasive (ett3 lecture, Sep 11 2026)

forest stand dynamics: disturbance opens pockets of light; stands initiate as a patch mosaic (ett3 lecture, Oct 2 2026)

SOURCES

Observations: Marko Kihara Manriquez, field visits Oct 2026. City facts: City of Cambridge, Alewife Reservation Constructed Wetland project overview and opening handout (Oct 15 2013); marsh zones from the City's project page. Construction sheets in hand (title, G-1, G-3, G-4) do not give the normal pool elevation.

Site Photos

PIN-06

Top: Fernanda Castillo's catalog photos (primary species). Bottom: keyframes from our scan walk of the bank, Oct 16 2026, matched to her species where they show them. IDs as in Fernanda Castillo's catalog.

FERNANDA'S PLANT CATALOG (primary species)

Japanese knotweed, Reynoutria japonica
invasive, prohibited in MA · Plant catalog: Fernanda Castillo

European buckthorn, Rhamnus cathartica
stems, understory · invasive · Plant catalog: Fernanda Castillo

European buckthorn, Rhamnus cathartica
black berries, bird dispersed · Plant catalog: Fernanda Castillo

Goldenrod, Solidago spp.
fall seed head · native · Plant catalog: Fernanda Castillo

Sugar maple, Acer saccharum
native · Plant catalog: Fernanda Castillo

Domestic apple, Malus domestica
fruit and leaves · Plant catalog: Fernanda Castillo

OUR BANK scan-walk keyframes, Oct 16 2026 (no separate field photos received)

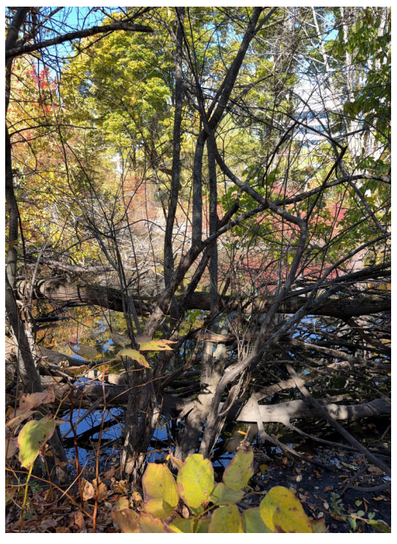
Buckthorn, multi-stem shrub, interior
matches Fernanda Castillo's European buckthorn · scan-walk keyframe, Oct 16

Goldenrod, ground layer, edge
matches Fernanda Castillo's goldenrod · scan-walk keyframe, Oct 16

Fallen utility pole on oaks
moss, branches, litter; pole 5.75 m (scan) · keyframe, Oct 16

Edge: woody stems thin out
herbs and vines on the slope; edge band s 5.5 to 6.75 m (scan)

Bank drop to standing water
0.43 m (1.4 ft) (scan); compacted bank, observed



Wetland margin: still, shallow water
fallen branches; about 0 to 1 ft deep (observed)

Nest · Habitat Feature

Two abandoned bird's nests from the bank, photographed in the studio. Habitat features layer (spec C1) and plastic reuse. Range of distances: wide (B), mid (A), close (C), plan (D).

PIN-07

A · mid: oblique, stone behind, the full branch (hero)

D · plan: top-down

Abandoned bird's nests (2) found on the bank, Alewife Brook Reservation, Oct 2026. Plastic wrappers woven into the structure. Collected and turned into a readymade sculpture.

Habitat features layer: nest 1 sat in a shrub about 6 ft up, about arm's length from the trunk, between stations 4 and 5 m on our transect (PIN-02, PIN-03); the second nest's location was not recorded.

Plastic reuse: the wrappers are woven in, not blown onto the nest, so the birds chose the material: adaptive reuse of plastic litter from the bank, possibly as weatherproofing. The same bank carries human use (an encampment, waste, plastic), so the litter is close at hand.

Photos: studio, Oct 29 2026 (A IMG_2582, B IMG_2559, C IMG_2543, D IMG_2549), from the originals. Light correction only: neutral white balance, contrast about +7 %, slight shadow lift; backdrop as shot. B trimmed top and bottom to fit; A, C, D full frame.

C · close: underside, woven plastic

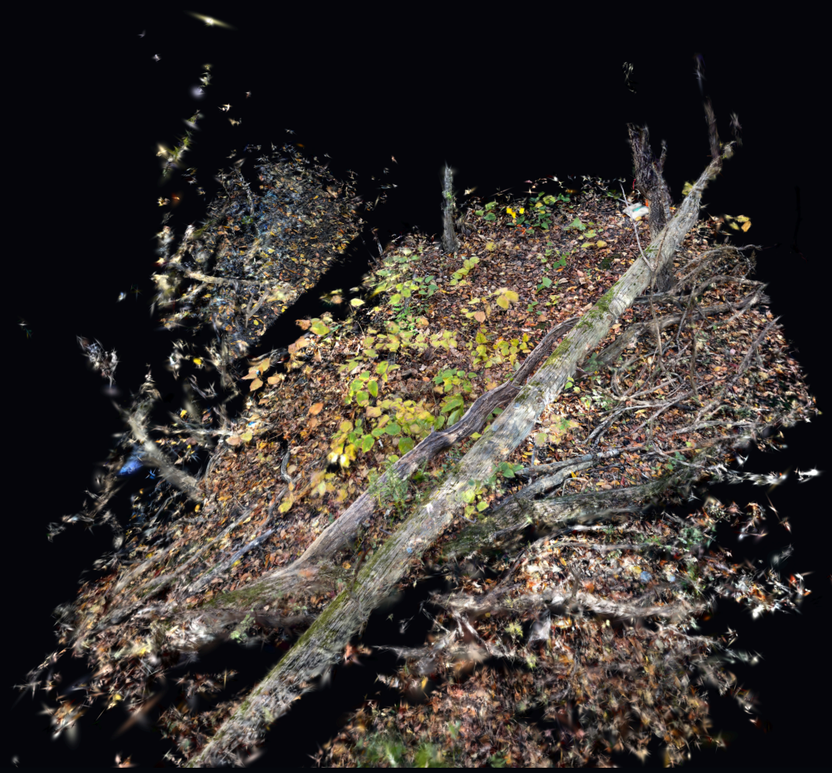
B · wide: front, nest on the branch, stone base

Scan + Splat Renders

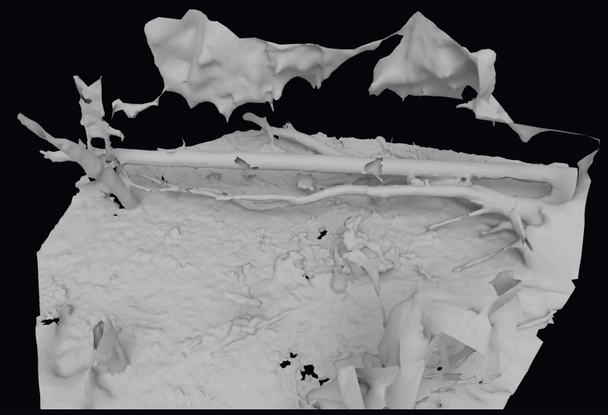
PIN-08

Gaussian splat (gsplat rasterizer) and Polycam mesh (Blender, trimmed) of the patch: the fallen utility pole on oaks, on ink.

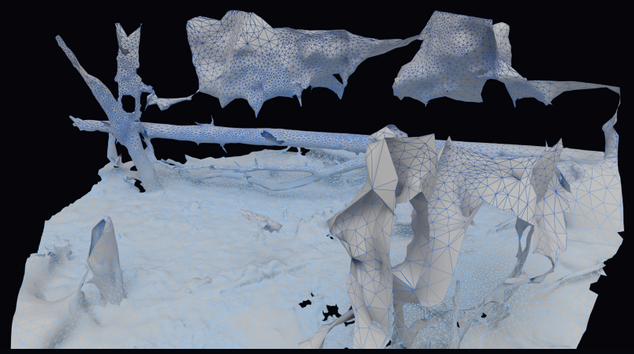
Splat, high orbit B: the patch as an island, pole diagonal across it · r 5.5 m, 5.0 m up, 24 mm



Mesh, clay, top-angled · 4.8 m out, 4.6 m up, 30 mm



Mesh wireframe, front-angled · 5.6 m out, 2.3 m up, 28 mm; wire blue by height



Splat, low close: the pole's long side · r 1.8 m, 28 mm

Splat, low close: moss on the pole · r 1.8 m, 28 mm

Splat, low close: down the length of the pole · r 1.8 m, 28 mm

Polycam scan; gaussian splat trained locally (COLMAP + Brush); renders: gsplat, Blender.

Animations (orbits, 360, walk-through): laserpirate.github.io/ett3/a3-groundtruth/animations/

Panels rendered at 600 ppi at their printed size. Scan and splat: Marko Kihara Manriquez, field visit Oct 16 2026.